Commutative Algebra

Shing Tak Lam

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Lecture 1

In this course, a ring is a commutative unital ring R. One non-commutative exception is the ring $\mathrm{End}(M)$, where M is an abelian group. This is a ring with pointwise addition, and composition as multiplication.

Definition 0.0.1 (module)
An *R-module* M is an abelian group M with an fixed ring homomorphism $\rho : R \to \mathrm{End}(M)$. We will write $r \cdot m := \rho(r)(m)$.

Remark 0.0.2. By definition, this implies that $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$, $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$ and $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$.

Example 0.0.3 (Examples of modules)
- Let k be a field. Then a k-module is the same as a k-vector space.
- Every abelian group is a $\mathbb{Z}$-module in a unique way, since we must have that $\rho(1) = \mathrm{id}_M$. Therefore, abelian groups and $\mathbb{Z}$-modules are the same thing.
- Every ring R is (trivially) an R-module.
- More generally, $R^{\oplus \mathbb{N}}$ (direct sum) and $R^{\mathbb{N}}$ (direct product) are R-modules.

1 Chain conditions

Definition 1.0.1 (Noetherian, Artinian module)
An R-module M is *Noetherian* if one of the following (equivalent) conditions hold:

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1. Every ascending chain of submodules $M_0 \subseteq M_1 \subseteq M_2 \subseteq \cdots$ stabilises. That is, it is eventually constant.
2. Every non-empty set Σ of submodules of M has a maximal element.

M is *Artinian* if we replace in the above: ascending with descending, maximal with minimal.

Lemma 1.0.2. An R-module M is Noetherian if and only if every submodule of M is finitely generated.

In particular, every Noetherian module is finitely generated. If $R = \mathbb{Z}[T_1, T_2, \ldots]$, with $M = R$ as an R-module. Then M is finitely generated. On the other hand, $M' = \langle T_1, T_2, T_3, \ldots \rangle$, is not finitely generated.

Definition 1.0.3 (Noetherian, Artinian ring)
A ring R is Noetherian (resp. Artinian) if it is Noetherian (resp. Artinian) as an R-module.

Example 1.0.4
1. $\mathbb{Z}$ is Noetherian (as it is a PID), but not Artinian (e.g. $\langle 2 \rangle \supseteq \langle 4 \rangle \supseteq \langle 8 \rangle \supseteq \cdots$).
2. $\mathbb{Z}[1/2]/\mathbb{Z}$ is Artinian, but not Noetherian as a $\mathbb{Z}$-module.
3. A ring R is Artinian if and only if R is Noetherian and R has Krull dimension 0.

Definition 1.0.5 (Exact sequence)
A sequence

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \cdots$$

of R-modules and R-module homomorphisms is *exact* if $\operatorname{im}(f_i) = \ker(f_{i+1})$ for all i.

Definition 1.0.6 (Short exact sequence)
A *short exact sequence* (SES) is an exact sequence of the form

$$0 \longrightarrow N \xhookrightarrow{\iota} M \twoheadrightarrow L \longrightarrow 0$$

That is, we have an embedding $\iota : N \hookrightarrow M$, and an isomorphism $L \cong M/\iota(N)$.

Lemma 1.0.7. Let

$$0 \longrightarrow N \longrightarrow M \longrightarrow L \longrightarrow 0$$

be an SES of R-modules. Then M is Noetherian (resp. Artinian) if and only if N and L are Noetherian (resp. Artinian).

Proof. We may assume without loss of generality that N is a submodule of M. Let $P_1 \subseteq P_2 \subseteq \ldots$ be an increasing (resp. decreasing) sequence of submodules of M. In this case,

$$N \cap P_1 \subseteq N \cap P_2 \subseteq \cdots$$

is an increasing (resp. decreasing) sequence of submodules of N, hence eventually constant. Similarly,

$$\frac{N + P_1}{N} \subseteq \frac{N + P_2}{N} \subseteq \cdots$$

is an increasing (resp. decreasing) sequence of submodules of $L = M/N$, hence eventually constant. For large n, we will have

$$P_n \subseteq P_{n+1} \quad N \cap P_n = N \cap P_{n+1} \quad N + P_n = N + P_{n+1}$$

Hence $P_n = P_{n+1}$ for large enough n. □

Corollary 1.0.8. If $M_1, \dots, M_n$ are Noetherian (resp. Artinian) R-modules, then $M_1 \oplus \cdots \oplus M_n$ is Noetherian (resp. Artinian).

Proof. By the lemma and induction. □

Recall a module homomorphism

$$\varphi : M_1 \oplus \cdots \oplus M_n \to L$$

is the same as a collection of module homomorphism $\varphi_i : M_i \to L$. This is also true for infinite direct sums (but not products!).

Proposition 1.0.9. For a Noetherian (resp. Artinian) ring R, every finitely generated R-module is Noetherian (resp. Artinian).

Proof. M is finitely generated if and only if there exists a surjection $R^n \twoheadrightarrow M$ for some $n \in \mathbb{N}$. The fact that R^n is Noetherian (resp. Artinian) implies that M is Noetherian (resp. Artinian), as quotients of Noetherian (resp. Artinian) modules are Noetherian (resp. Artinian). □

Definition 1.0.10 (algebra)

An *R-algebra* A is a ring A with a fixed ring homomorphism $\rho : R \to A$. We will write $r \cdot a := \rho(r)a$.

Definition 1.0.11 (noetherian algebra)

An R-algebra is *Noetherian* if it is Noetherian as a ring.

Remark 1.0.12. Every R-algebra is an R-module.

Example 1.0.13

The polynomial ring $k[T_1, \dots, T_n]$ is a k-algebra. Do note however that it is a finitely generated by $T_1, \dots, T_n$ as a k-algebra, but it is infinite dimensional as a k-vector space.

Definition 1.0.14 (algebra homomorphism)

$\varphi : A \to B$ is an *R-algebra homomorphism* if φ is a ring homomorphism and $\varphi(r \cdot 1_A) = r \cdot 1_B$.

Definition 1.0.15 (finitely generated algebra)

An R-algebra A is *finitely generated* if there exists a surjective R-algebra homomorphism $R[T_1, \dots, T_n] \twoheadrightarrow A$ for some $n \in \mathbb{N}$.

Lecture 2

Theorem 1.0.16 (Hilbert basis theorem)**.** Every finitely generated algebra A over a Noetherian ring R is Noetherian (as a ring).

For example, if k is a field, then $k[T_1, \dots, T_n]$ is Noetherian.

Proof. It suffices to prove for $A = R[T_1, \dots, T_n]$, since every finitely generated algebra is a quotient of $R[T_1, \dots, T_n]$. Moreover, by induction, suffices to prove the result for $A = R[T]$.

Let $\mathfrak{a}$ be an ideal of $A = R[T]$. For every $i \geq 0$, define

$$\mathfrak{a}(i) = \left\{c_0 \mid c_0 t^i + \cdots + c_i t^0 \in \mathfrak{a}\right\}$$

for the set of all leading coeffients of elements of degree i in $\mathfrak{a}$ (and containing 0). In this case, $\mathfrak{a}(i) \subseteq R$ is an ideal, and we have an ascending chain of ideals

$$\mathfrak{a}(i) \subseteq \mathfrak{a}(i+1) \subseteq \cdots$$

Since R is Noetherian, each $\mathfrak{a}$ is finitely generated (as an ideal), and the ascending sequence of ideal stabilises. That is,

$$\mathfrak{a}(m') = \mathfrak{a}(m)$$

for all $m' \geq m$. We write $\mathfrak{a}(i) = \langle b_{i,1}, \dots, b_{i,m_i} \rangle$, where $b_{i,j} \in R$. Let $f_{i,j} \in \mathfrak{a}$ be a polynomial of degree i, with leading coefficient $b_{i,j}$. Define the new ideal

$$\mathfrak{b} = \left\langle f_{i,j} \mid i \leq m, 1 \leq j \leq m_i \right\rangle \trianglelefteq R[T]$$

In this case, $\mathfrak{b}(i) = \mathfrak{a}(i)$ for all i. By construction, $\mathfrak{b} \subseteq \mathfrak{a}$.

Suppose for contradiction that $\mathfrak{a} \nsubseteq \mathfrak{b}$. Take $f \in \mathfrak{a} \setminus \mathfrak{b}$ of minimal degree i. But $\mathfrak{b}(i) = \mathfrak{a}(i)$, and so there exists $g \in \mathfrak{b}$, of degree i, and with the same leading coefficient as f. That is, $\deg(f - g) < i$. By minimality, $f - g \in \mathfrak{b}$, and so $f = (f - g) + g \in \mathfrak{b}$. Contradiction. □

Therefore, if we have a subset $S \subseteq R[T_1, \dots, T_n]/I$, then $\langle S \rangle = \langle S_0 \rangle$, where $S_0 \subseteq S$ is finite.

2 Tensor products

Let M, N be R-modules. An informal definition of their tensor product is

$$M \otimes_R N = \left\{ \sum_{i=1}^{\ell} m_i \otimes n_i \;\middle|\; m_i \in M, n_i \in N \right\}$$

where we have the relations $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$, $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$, and that for $r \in R$, $(rm) \otimes n = r(m \otimes n) = m \otimes (rn)$.

For example, consider $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3$. Then

$$x \otimes y = (3x) \otimes y = x \otimes (3y) = x \otimes 0 = 0$$

and so, $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3 = 0$. On the other hand, if we have vector spaces, then

$$\mathbb{R}^m \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong \mathbb{R}^{m\ell}$$

Recall $f : M \times N \to L$ is *R-bilinear* if $n \mapsto f(m_0, n)$ and $m \mapsto f(m, n_0)$ are R-linear for all $m_0 \in M, n_0 \in N$.

Definition 2.0.1 (tensor product of modules)

Let M, N be R-modules, let

$$\mathcal{F} = R^{\oplus(M \times N)} = \operatorname{span}_R \left\{ e_{(m,n)} \mid m \in m, n \in N \right\}$$

be the free module indexed by $m \times n$, and define $\mathcal{K} \subseteq \mathcal{F}$ for the submodule generated by the relations

(where we write (m, n) for $e_{(m,n)}$)

$$\begin{aligned}(m, n_1) + (m, n_2) &= (m, n_1 + n_2)\\ (m_1, n) + (m_2, n) &= (m_1 + m_2, n)\\ r(m, n) &= (rm, n)\\ r(m, n) &= (m, rn)\end{aligned}$$

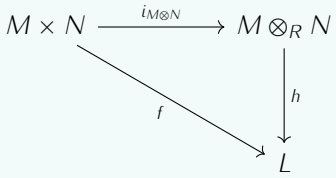
The *tensor product* is

$$M \otimes_R N := \frac{\mathcal{F}}{\mathcal{K}}$$

We have an R-bilinear map

$$\begin{aligned}i_{M\otimes N} : M \times N &\to M \otimes_R N\\ (m, n) &\mapsto m \otimes n\end{aligned}$$

Proposition 2.0.2 (universal property of tensor product)**.** For every R-module L and any R-bilinear map $f : M \times N \to L$, there exists a unique R-linear $h : M \otimes N \to L$, making the diagram



commute.

Proof. Uniqueness is clear, since we must have that

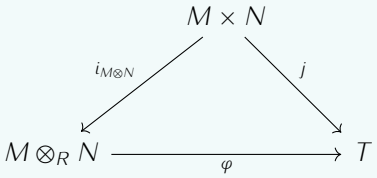
$$h(m \otimes n) = f(m, n)$$

since the pure tensors generate, h must be unique, if it exists. Therefore, suffices to show the above extends to an R-linear map $M \otimes_R N \to L$. This follows from the map

$$\begin{aligned}R^{\oplus(M\times N)} &\to L\\ e_{(m,n)} &\mapsto f(m, n)\end{aligned}$$

extending to a linear map (by the universal property of the direct sum), and that this map vanishes on $\mathcal{K}$. Therefore, h extends to $M \otimes_R N$ from the pure tensors. □

Proposition 2.0.3. Let M, N be R-modules, T an R-module, $j : M \times N \to T$ an R-bilinear map, (T, j) satisfying the universal property of tensors. Then there exists a unique R-linear isomorphism $\varphi : M \otimes N \to T$, such that



commutes.

Proof. By the universal property of tensor product, such a map φ exists, with $\varphi(m \otimes n) = j(m, n)$. Similarly, we have a homomorphism $\psi : T \to M \otimes_R N$. In particular,

$$\psi \circ \varphi \circ i_{M\otimes N} = i_{M\otimes N} = \mathrm{id}_{M\otimes N} \circ i_{M\otimes N}$$

In particular, by uniqueness in the universal property, we must have that $\psi \circ \varphi = \mathrm{id}_{M\otimes N}$. □

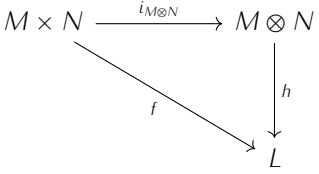
Proposition 2.0.4. Suppose M, N are R-modules, then

$$\sum_i m_i \otimes n_i = 0 \in M \otimes_R N$$

if and only if for all R-modules L, and every R-bilinear map $f : M \times N \to L$ has

$$\sum_i f(m_i, n_i) = 0$$

Proof. Suppose $\sum m_i \otimes n_i = 0$, let $f : M \times N \to L$ be bilinear. Then f factors through $M \times N \to M \otimes_R N$, and we can write



In this case, we have that

$$\sum_i f(m_i, n_i) = \sum_i h(i(m, n)) = \sum_i h(m_i \otimes n_i) = h\left(\sum_i m_i \otimes n_i\right) = h(0) = 0$$

Conversly, if

$$\sum_i m_i \otimes n_i \neq 0$$

then by definition,

$$\sum_i i_{m\otimes n}(m_i, n_i) \neq 0$$

□

Example 2.0.5

Let k be a field, and consider the tensor product

$$k^m \otimes k^\ell$$

Suppose k^m has basis $\{e_1, \ldots, e_m\}$, and k^ℓ has basis $\{f_1, \ldots, f_\ell\}$, then

$$k^m \otimes k^\ell = \operatorname{span}_k\{v \otimes w \mid v \in k^m, w \in k^\ell\} = \operatorname{span}_k\left\{e_i \otimes f_j\right\}$$

Claim 2.0.6. $\{e_i \otimes f_j\}$ is a basis.

Proof. Suppose we have

$$\sum_{ij} \alpha_{ij}(e_i \otimes f_j) = 0$$

For every $1 \leq a \leq m, 1 \leq b \leq \ell$, define a bilinear map

$$T_{ab} : k^m \times k^\ell \to k$$
$$T_{ab}((v_i), (w_j)) = v_a w_b$$

This is a k-bilinear map. By proposition 2.0.4,

$$0 = \sum_{i,j} \alpha_{ij} T_{ab}(e_i, f_j) = \sum_{i,j} \alpha_{ij} \delta_{ia} \delta_{jb} = \alpha_{ab}$$

□

Example 2.0.7

More concretely, let us consider $\mathbb{R}^2 \otimes \mathbb{R}^2$. We have a basis of size 4, given by

$$e_1 \otimes f_1, e_1 \otimes f_2, e_2 \otimes f_1, e_2 \otimes f_2$$

What do pure tensors look like?

$$(\alpha e_1 + \beta e_2) \otimes (\gamma f_1 + \delta f_2) = \alpha\gamma(e_1 \otimes f_1) + \alpha\delta(e_1 \otimes f_2) + \beta\gamma(e_2 \otimes f_1) + \beta\delta(e_2 \otimes f_2)$$

These are not generic elements of $\mathbb{R}^2 \otimes \mathbb{R}^2$, since the vectors

$$(\alpha\gamma, \alpha\delta) \quad \text{and} \quad (\beta\gamma, \beta\delta)$$

are linearly dependent. In particular,

$$e_1 \otimes f_1 + 2e_1 \otimes f_2 + 3e_2 \otimes f_1 + 4e_2 \otimes f_2$$

is *not* a pure tensor.

Example 2.0.8 (warning)

First consider

$$\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

In this case,

$$2 \otimes 1 = 1 \otimes 2 = 1 \otimes 0 = 0$$

Now consider

$$2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

But in this case,

$$2 \otimes 1 \neq 0$$

since we can define a bilinear map

$$B : 2\mathbb{Z} \times \mathbb{Z}/2 \to \mathbb{Z}/2$$
$$B(2m, x) = mx$$

In this case,

$$B(2, 1) = 1 \cdot 1 = 1 \neq 0$$

However, if $M' \leq M, N' \leq N$ are submodules, and

$$\sum_i m_i \otimes n_i = 0$$

in $M' \otimes N'$, then

$$\sum_i m_i \otimes n_i = 0$$

in $M \otimes N$.

Proposition 2.0.9. If

$$\sum m_i \otimes n_i = 0 \in M \otimes_R N$$

then there are finitely generated R-submodules $M' \leq M, N' \leq N$, such that

$$\sum m_i \otimes n_i = 0 \in M' \otimes_R N'$$

Intuitively, a proof that the sum is zero is finite, and so it can only involve finitely many expressions. We can take them to be the generators.

Proof.

$$\sum m_i \otimes n_i 0 = \in M \otimes N = \frac{R^{\oplus(M\times N)}}{\mathcal{K}}$$

then

$$\sum_i e_{(m_i,n_i)} = 0 \in \mathcal{K}$$

This means that we can write the left hand side as a finite sum of the generators of $\mathcal{K}$. Taking all the elements of M and N which appear, gives the result. □

Corollary 2.0.10. Let A, B be torsion-free abelian groups, then $A \otimes_{\mathbb{Z}} B$ is torsion free.

Proof. Suppose

$$n \cdot \left(\sum_i a_i \otimes b_i \right) = 0 \in A \otimes B$$

for some $n \geq 1$. By proposition 2.0.9, there exists finitely generated subgroups $A' \leq A, B' \leq B$, such that

$$n \cdot \left(\sum_i a_i \otimes b_i \right) = 0 \in A' \otimes B'$$

By the structure theorem of finitely generated abelian groups, $A' \cong \mathbb{Z}^r, B' \cong \mathbb{Z}^s$, and so we have that

$$A' \otimes B' \cong \mathbb{Z}^{rs}$$

which is torsion free. Contradiction. □

Example 2.0.11

$$\mathbb{C}^2 \otimes_{\mathbb{C}} \mathbb{C}^3 \cong \mathbb{C}^6$$

as $\mathbb{C}$-vector spaces, and we also have that $\mathbb{C}^6 \cong \mathbb{R}^12$ as $\mathbb{R}$-vector spaces. On the other hand,

$$\mathbb{C}^2 \otimes_{\mathbb{R}} \mathbb{C}^3 \cong \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{R}^6 \cong \mathbb{R}^{24}$$

Proposition 2.0.12.

1. $M \otimes N \cong N \otimes N$
2. $(M \otimes N) \otimes P \cong M \otimes (N \otimes P) \cong M \otimes N \otimes P$, where we define $M \otimes N \otimes P$ using trilinear maps.
3. $\left(\bigoplus_i M_i\right) \otimes P \cong \bigoplus_i (M_i \otimes P)$
4. $R \otimes_R M \cong M$,

Proof. See examples sheet 1. □

Example 2.0.13

Using proposition 2.0.12, we can compute

$$\begin{aligned} R^m \otimes R^\ell &\cong \left(\oplus_{i=1}^m R\right) \otimes \left(\oplus_{j=1}^\ell R\right) \\ &\cong \oplus_{i,j} R \\ &\cong R^{m\ell} \end{aligned}$$

2.1 Tensor product of R-linear maps

Proposition 2.1.1. For R-linear maps $f : M \to M', g : N \to N'$, then there exists a unique R-linear map

$$f \otimes g : M \otimes N \to M' \otimes N'$$

with

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

Proof. Uniqueness is clear since the pure tensors generate. For existence, we can use the universal property on the R-bilinear map

$$\begin{aligned} T : M \times N &\to M' \otimes N' \\ T(m, n) &= f(m) \otimes g(n) \end{aligned}$$

□

Lecture 4

Exercise: $(f \otimes g) \circ (h \otimes i) = (f \circ h) \otimes (h \circ i)$.

Example 2.1.2

Let $T : k^a \to k^c$ and $S : k^b \to k^d$ be linear. Then

$$(T \otimes S)(e_i \otimes e_j) = T(e_i) \otimes S(e_j) = \sum_{\ell,t} [T]_{\ell i}[S]_{tj}(f_\ell \otimes f_t)$$

where $[T]$ is the matrix representation of T. If we order the basis of $k^a \otimes k^b$ by

$$e_1 \otimes e_1, \ldots, e_1 \otimes e_c, e_2 \otimes e_1, \ldots, e_2 \otimes e_c, \ldots, e_a \otimes e_c$$

and a similar ordering for the range, then

$$[T \otimes S] = \begin{pmatrix} [T]_{11}S & \cdots & [T]_{1a}S \\ \vdots & \ddots & \vdots \\ [T]_{c1}S & \cdots & [T]_{ca}S \end{pmatrix}$$

is the *Kronecker product* of $[T]$ and $[S]$.

Proposition 2.1.3. Let $f : M \to M', g : N \to N'$ be $\mathbb{R}$-linear.

(i) If f, g are isomorphisms, then so is $f \otimes g$,

(ii) if f and g are surjective, so is $f \otimes g$.

Proof. For (i), $(f^{-1} \otimes g^{-1}) = (f \otimes g)^{-1}$, since we have that $(f \otimes g) \circ (h \otimes i) = (f \circ h) \otimes (h \circ i)$.
For (ii), notice that $\mathrm{im}(f \otimes g)$ contains all pure tensors in $M' \otimes N'$. □

Example 2.1.4

If $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = pn$, then we have

$$(f \otimes \mathrm{id}) : \mathbb{Z} \otimes \mathbb{Z}/p \to \mathbb{Z} \otimes \mathbb{Z}/p$$

is the zero map, as

$$(f \otimes \mathrm{id})(a \otimes b) = (pa) \otimes b = a \otimes (pb) = a \otimes 0 = 0$$

But $\mathbb{Z} \otimes \mathbb{Z}/p \cong \mathbb{Z}/p$ which is nonzero.

2.2 Tensor product of algebras

Let B, C be R-algebras. Then we have $B \otimes_R C$ as an R-module. We would like to define the multiplication by

$$(b \otimes c)(b' \otimes c') = (bb') \otimes (cc')$$

This is well-defined. Fix $(b, c) \in B \times \mathbb{C}$, then we have a bilinear map

$$\begin{aligned} B \times C &\to B \otimes C \\ (b', c') &\mapsto (bb') \otimes (cc') \end{aligned}$$

which gives us a map $B \otimes C \to B' \otimes C'$, with

$$b' \otimes c' \mapsto (bb') \otimes (cc')$$

It is easy to show that this then satisfies the ring axioms. Hence $B \otimes C$ is a ring. The R-algebra structure will be given by

$$\begin{aligned} R &\to B \otimes C \\ r &\mapsto (r1_B) \otimes 1_C = r(1_B \otimes 1_C) = 1_B \otimes (r1_C) \end{aligned}$$

Example 2.2.1

There is an isomorphism

$$\varphi : R[x_1, \ldots, x_n] \otimes_R R[t_1, \ldots, t_r] \cong R[x_1, \ldots, x_n, t_1, \ldots, t_r]$$

Proof. We have an R-basis for the left hand side, which is

$$x^k \otimes t^\ell$$

and we also have a R-basis for the right hand side,

$$x^k t^\ell$$

Define

$$\varphi(x^k \otimes t^\ell) = x^k t^\ell$$

which gives us a R-module isomorphism. Moreover,

$$\varphi(r \otimes 1) = r1 = 1$$

and by distributivity, suffices to show

$$\varphi((x^k \otimes t^\ell)(x^m \otimes t^n)) = x^k t^\ell x^m t^n$$

which is clear by definition. □

More generally,

$$\frac{R[x_1,\dots,x_n]}{I}\otimes\frac{R[t_1,\dots,t_r]}{J}\cong\frac{R[x_1,\dots,x_n,t_1,\dots,t_r]}{L}\cong\frac{R[x_1,\dots,x_n,t_1,\dots,t_r]}{I^e+J^e}$$

where $I^e=\langle I\rangle\trianglelefteq R[x_1,\dots,x_n,t_1,\dots,t_r]$ denotes the extension of I.

Example 2.2.2

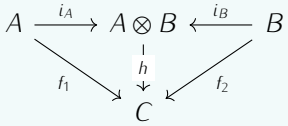
$\frac{\mathbb{C}[x,y,z]}{\langle f,g\rangle}\otimes\frac{\mathbb{C}[w,u]}{h}$ is isomorphic as $\mathbb{C}$-algebras to

$$\frac{\mathbb{C}[x,y,z,w,u]}{\langle f,g,h\rangle}$$

Proposition 2.2.3 (universal property of tensor product of algebras). Let A,B be R-algebras, for every R-algebra C, and R-algebra homomorphisms $f_1:A\to C$ and $f_2:B\to C$, there exists a unique R-algebra map

$$h:A\otimes B\to C$$

such that



commutes, where $i_A(a)=a\otimes 1$, $i_B(b)=1\otimes b$. Moreover, this characterises $(A\otimes B,i_A,i_B)$ uniquely (up to isomorphism).

Proof. $A\otimes B$ is generated, as an R-algebra, by

$$\{a\otimes 1\mid a\in A\}\cup\{1\otimes b\mid b\in B\}$$

This then implies the uniqueness of h, as it defines h on the generators. For the existence, define the bilinear map $A\times B\to C$, given by

$$f(a,b)=f_1(a)f_2(b)$$

Using the universal property of tensor product of modules, there exists $h:A\otimes B\to C$ which is R-linear, with

$$h(a\otimes b)=f_1(a)f_2(b)$$

It is then easy to show that h is an algebra homomorphism. □

Consider $R[x_1,\dots,x_n,t_1,\dots,t_r]$ from above. We have natural embeddings from $R[x_1,\dots,x_n]$ and $R[t_1,\dots,t_n]$. Given f_1,f_2 as above, we see that the image of the x_i is determined by f_1, and the image of t_i is determined by f_2. Therefore,

$$R[x_1,\dots,x_n,t_1,\dots,t_r]\cong\mathbb{R}[x_1,\dots,x_n]\otimes R[t_1,\dots,t_r]$$

as it satisfies the universal property.

Lecture 5

If we have $f:A\to A'$, $g:B\to B'$ which are algebra homomorphisms, then the tensor product of R-linear maps,

$$f\otimes g:A\otimes B\to A'\otimes B'$$

is an R-algebra homomorphism. Moreover, we have R-algebra isomorphisms

- $(R/I)\otimes(R/J)\cong R/(I+J)$
- $A\otimes B\cong B\otimes A$,
- $(A\otimes B)\otimes C\cong A\otimes(B\otimes C)$,
- $A\otimes B^n\cong(A\otimes B)^n$,

2.3 Restriction and extension of scalars

Restriction of scalars

We will have a ring homomorphisms $f : R \to S$, let M be an S-module, so M is also an R-module,

$$r \cdot m := f(r)m$$

for $r \in R, m \in M$. The fact that this is a module is clear by our definition, since it is just the composition

$$R \xrightarrow{\quad f \quad} S \longrightarrow \mathrm{End}(M)$$

> **Example 2.3.1**
> If we consider the embedding $\mathbb{R} \hookrightarrow \mathbb{C}$, then $\mathbb{C}^n$ is a C-vector space, but also an $\mathbb{R}$-vector space, of dimension $2n$.

Extension of scalars

Let $f : R \to S$ be a ring homomorphism, M be an S-module (thus an R-module by restriction of scalars), N is an R-module. From this, we can form

$$M \otimes_R N$$

which is an R-module. In fact, $M \otimes_R N$ is also an S-module, with

$$s \cdot (m \otimes n) := (sm) \otimes n$$

Is this well defined? We have an R-bilinear map

$$\begin{aligned} M \times N &\to M \otimes_R N \\ (m, n) &\mapsto (sm) \otimes n \end{aligned}$$

By the universal property, we have a map

$$h_s : M \otimes_R N \to M \otimes_R N$$

which is R-linear, and $h_s(m \otimes n) = (sm) \otimes n$. Now define

$$\begin{aligned} \varphi : S &\to \mathrm{End}(M \otimes_R N) \\ \varphi(s) &= h_s \end{aligned}$$

Which is a ring homomorphism, and so, we have an S-module structure on $M \otimes_R N$.

> **Example 2.3.2**
> We know from before that $S \otimes_R R \cong S$ as R-module, with
>
> $$s \otimes r \mapsto s \cdot f(r)$$
>
> But in fact, this is also S-linear, since
>
> $$s' \cdot (s \otimes r) = (s's) \otimes r \mapsto s's \cdot f(r)$$
>
> For example, this implies that
>
> $$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R} \cong \mathbb{C}$$
>
> as $\mathbb{C}$-vector spaces.

Example 2.3.3

If M is an S-module, N_i are R-modules, then

$$M \otimes_R \left(\bigoplus_i N_i \right) \cong \bigoplus_i (M \otimes_R N_i)$$

as S-modules.

In this case,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^n \cong \mathbb{C}^n$$

as $\mathbb{C}$-vector spaces.

Example 2.3.4

Consider $\mathbb{C}^n$ as a $\mathbb{C}$-module. Restricting to $\mathbb{R}$,

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

as $\mathbb{R}$-vector spaces. Now extending scalars,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^{2n} \cong \mathbb{C}^{2n}$$

as $\mathbb{C}$-vector spaces.

Example 2.3.5

Now consider $\mathbb{R}^n$ as an $\mathbb{R}$-vector space. Extending scalars,

$$\mathbb{R}^n \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}^n$$

over $\mathbb{C}$. Restricting to $\mathbb{R}$,

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

Example 2.3.6

Consider $\mathbb{Z}^n$ as an $\mathbb{Z}$-module, and let $f : \mathbb{Z} \to \mathbb{Z}/2$ be the quotient map. Extending scalars,

$$(\mathbb{Z}/2) \otimes_{\mathbb{Z}} \mathbb{Z}^n \cong (\mathbb{Z}/2)^n$$

Example 2.3.7

Consider

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell$$

One way to compute this:

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong_{\mathbb{R}} \mathbb{R}^{2n} \otimes \mathbb{R}^\ell \cong_{\mathbb{R}} \mathbb{R}^{2n\ell} \cong_{\mathbb{R}} \mathbb{C}^{n\ell}$$

where $\cong_{\mathbb{R}}$ denotes isomorphism as $\mathbb{R}$-vector spaces. Another way to do this:

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong_{\mathbb{C}} \mathbb{C}^n \otimes_{\mathbb{C}} \left(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^\ell \right) \cong_{\mathbb{C}} \mathbb{C}^n \otimes \mathbb{C}^\ell \cong_{\mathbb{C}} \mathbb{C}^{n\ell}$$

The first isomorphism is given by

$$v \otimes u \mapsto v \otimes (1 \otimes u)$$

Combining these, the isomorphism $\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \to \mathbb{C}^n \otimes \mathbb{C}^\ell$ sends

$$v \otimes u \mapsto v \otimes u$$

where we use the inclusion $\mathbb{R}^\ell \hookrightarrow \mathbb{C}^\ell$.

Proposition 2.3.8. Let M be an S-module, N be an R-module, then

$$M \otimes_R N \cong M \otimes_S (S \otimes_R N)$$

as S-modules. In particular, the isomorphism is given by

$$\begin{aligned} m \otimes n &\mapsto m \otimes (1 \otimes n) \\ (sm) \otimes n &\leftarrow\!\shortmid m \otimes (s \otimes n) \end{aligned}$$

Proposition 2.3.9. Let M, M' be S-modules, N, N' be R-modules, then we have S-module isomorphisms

(i) $M \otimes_R N \cong N \otimes_R M$, via $m \otimes n \to n \otimes m$

(ii) $(M \otimes_R N) \otimes_R N' \cong M \otimes_R (N \otimes_R N')$

(iii) $(M \otimes_R N) \otimes_S M' \cong M \otimes_S (N \otimes_R M')$

(iv) $M \otimes_R (\bigoplus_i N_i) \cong \bigoplus_i (M \otimes_R N_i)$

Proof. We will prove (iii). Using proposition 2.3.8, we have

$$\begin{aligned} (M \otimes_R N) \otimes_S M' &\cong (M \otimes_S (N \otimes_R S)) \otimes_S M' \\ &\cong M \otimes_S \left((N \otimes_R S) \otimes_S M' \right) \\ &\cong M \otimes_S (N \otimes_R M') \end{aligned}$$

□

Example 2.3.10

As $\mathbb{C}$-vector spaces,

$$\mathbb{C} \otimes_{\mathbb{R}} (\mathbb{R}^\ell \otimes_{\mathbb{R}} \mathbb{R}^k) \cong (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^\ell) \otimes_{\mathbb{C}} (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^k) \cong \mathbb{C}^\ell \otimes \mathbb{C}^k \cong \mathbb{C}^{\ell k}$$

Corollary 2.3.11. If N, N' are R-modules, then

$$S \otimes_R (N \otimes N') \cong_S (S \otimes_R N) \otimes_S (S \otimes N')$$

Proof. By proposition 2.3.8 and proposition 2.3.9 (ii):

$$S \otimes_R (N \otimes_{\mathbb{R}} N') \cong (S \otimes_R N) \otimes_R N' \cong (S \otimes_R N) \otimes_S (S \otimes_R N')$$

□

Lecture 6

By induction, we have that

$$S \otimes_R (N_1 \otimes_R \cdots \otimes_R N_\ell) \cong (S \otimes_R N_1) \otimes_S \cdots \otimes_S (S \otimes_R N_1)$$

Extension of scalars for morphisms

Let $f : N \to N'$ be R-linear, where N, N' are R-modules, M is an S-module. Then we have a map

$$\mathrm{id} \otimes f : M \otimes_R N \to M \otimes_R N'$$

In particular, it is S-linear, as

$$(\mathrm{id}\otimes f)(s(m\otimes n)) = (\mathrm{id}\otimes f)((sm)\otimes n) = (sm)\otimes f(n) = s(m\otimes f(n)) = s(\mathrm{id}\otimes f)(m\otimes n)$$

Given $T:\mathbb{R}^n\to\mathbb{R}^\ell$ which is an $\mathbb{R}$-linear map, $\mathbb{R}^n$ with basis $e_1,\dots,e_n$ and $\mathbb{R}^\ell$ with basis $f_1,\dots,f_\ell$. In this case, consider

$$\mathrm{id}\otimes T : \mathbb{C}\otimes\mathbb{R}^n \to \mathbb{C}\otimes\mathbb{R}^\ell$$

Note that $\mathbb{C}\otimes\mathbb{R}^n$ has basis $1\otimes e_1,\dots,1\otimes e_n$. In particular,

$$(\mathrm{id}\otimes T)(1\otimes e_i) = 1\otimes T(e_i) = 1\otimes\sum_{j=1}^{\ell} T_{ji}f_j = \sum_{j=1}^{\ell} T_{ji}(1\otimes f_j)$$

Thus, T and $\mathrm{id}\otimes T$ have the same matrix representation.

Extension of scalars of algebras

Let A, B be R-algebras. Recall that in this case, $A\otimes_R B$ is also an R-algebra. In fact, $A\otimes_R B$ is an A-algebra (and by symmetry a B-algebra). For example, we have

$$\begin{aligned} A &\to A\otimes_R B\\ a &\mapsto a\otimes 1\end{aligned}$$

Example 2.3.12

$S\otimes_R R[x_1,\dots,x_n]\cong_S S[x_1,\dots,x_n]$ (where $\cong_S$ denotes isomorphism of S-algebras).

Proof. We already have an S-module isomorphism

$$\varphi : S\otimes_R R[x_1,\dots,x_n]\to S[x_1,\dots,x_n]$$

with $\varphi(s\otimes f) = sf$. It is easy to show that

$$\varphi(s\otimes 1) = s$$

and that φ preserves multiplication. □

More generally, we have that

$$S\otimes\left(\frac{R[x_1,\dots,x_n]}{I}\right)\cong\frac{S[x_1,\dots,x_n]}{I^e}$$

where $I^e = \langle f(I)\rangle$ is the ideal generated by I under the ring homomorphism $f:R\to S$.

Proposition 2.3.13. Suppose A is an R-algebra, B is an S-algebra, then $A\otimes_R B$ is an S-algebra. Moreover,

$$A\otimes_R B\cong_{S\text{-alg}}(A\otimes_R S)\otimes B$$

Proof. $A\otimes_R B$ is a B-algebra, and we can then restrict scalats to S. The isomorphism is clear from the module case, as all we need to check is it preserves multiplication. □

Proposition 2.3.14. Suppose A, B are R-algebras, then

$$S\otimes_R(A\otimes_R B)\cong_{S\text{-alg}}(S\otimes_R A)\otimes_S(S\otimes_R B)$$

2.4 Exactness properties of the tensor product

Let M be a fixed R-module. Define

$$\mathrm{T}_M(N) = M \otimes_R N$$

where N is an R-module. If $f : N \to N'$ is R-linear, then we have an induced map

$$\mathrm{T}_M(f) = \mathrm{id}_M \otimes f : \mathrm{T}_M(N) \to \mathrm{T}_M(N')$$

Suppose we have an exact sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

of R-modules. We will show that we have an exact sequence

$$\mathrm{T}_M(A) \xrightarrow{\mathrm{T}_M(f)} \mathrm{T}_M(B) \xrightarrow{\mathrm{T}_M(g)} \mathrm{T}_M(C) \longrightarrow 0$$

That is, T_M is a *right exact functor* from R-modules to R-modules.

Definition 2.4.1 (Hom)
Suppose Q, P are R-modules, then we can define

$$\mathrm{Hom}_R(Q, P) = \{f : Q \to P \mid f \text{ is } R\text{-linear}\}$$

This is an R-module itself, with

$$(r \cdot \varphi)(q) = r \cdot \varphi(q)$$

Definition 2.4.2 (Hom functors)
We have two functors,

1. $\mathrm{Hom}_R(Q, \cdot)$, where Q is a fixed R-module,
2. $\mathrm{Hom}_R(\cdot, P)$, where P is a fixed R-module.

Suppose we have $f : N \to N'$ which is R-linear, then the action on morphisms are

$$\begin{aligned} \mathrm{Hom}_R(Q, f) : \mathrm{Hom}_R(Q, N) &\to \mathrm{Hom}_R(Q, N') \\ \varphi &\mapsto f \circ \varphi =: f_*(\varphi) \end{aligned}$$

On the other hand, $\mathrm{Hom}_R(\cdot, P)$ is contravariant. That is,

$$\begin{aligned} \mathrm{Hom}_R(f, P) : \mathrm{Hom}_R(N', P) &\to \mathrm{Hom}_R(N, P) \\ \varphi &\mapsto \varphi \circ f =: f^*(\varphi) \end{aligned}$$

Proposition 2.4.3 (left exactness of the Hom-functors).

1. If

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

is exact, then so is

$$0 \longrightarrow \mathrm{Hom}_R(Q, A) \xrightarrow{\mathrm{Hom}_R(Q,f)} \mathrm{Hom}_R(Q, B) \xrightarrow{\mathrm{Hom}_R(Q,g)} \mathrm{Hom}_R(Q, C)$$

2. If

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact, then so is

$$0 \longrightarrow \mathrm{Hom}_R(C,P) \xrightarrow{\mathrm{Hom}_R(g,P)} \mathrm{Hom}_R(B,P) \xrightarrow{\mathrm{Hom}_R(f,P)} \mathrm{Hom}_R(A,P)$$

In both cases, we say that the respective Hom functor is *left exact*.

Proof. Omitted. □

Lemma 2.4.4. Consider a (not necessarily exact) sequence

$$A \xrightarrow{f} B \xrightarrow{g} C$$

and suppose for all R-module P, the sequence

$$\mathrm{Hom}_R(C,P) \longrightarrow \mathrm{Hom}_R(B,P) \longrightarrow \mathrm{Hom}_R(A,P)$$

is exact, then the original sequence is exact.

Proof. **Step 1:** let $P = C$. Then we get the sequence

$$\mathrm{Hom}_R(C,C) \longrightarrow \mathrm{Hom}_R(B,C) \longrightarrow \mathrm{Hom}_R(A,C)$$

which is exact by assumption. Under this,

$$\mathrm{id}_C \mapsto \mathrm{id}_C \circ g = g \mapsto g \circ f$$

Thus, we have that $g \circ f = 0$, and so $\mathrm{im}(f) \subseteq \ker(g)$.

Step 2: Let $P = \mathrm{coker}\, f = \frac{B}{\mathrm{im}(f)}$. In this case, we have

$$\mathrm{Hom}(C, \mathrm{coker}(f)) \longrightarrow \mathrm{Hom}(B, \mathrm{coker}(f)) \longrightarrow \mathrm{Hom}(A, \mathrm{coker}(f))$$

Let $h : B \to \mathrm{coker}(f)$ denote the quotient map. Then $h \circ f = 0$, and so by exactness, there exists $e : C \to \mathrm{coker}(f)$, with

$$\mathrm{Hom}(g, \mathrm{coker}(f))(e) = e \circ g = h$$

In particular, $\ker(g) \subseteq \ker(h) = \mathrm{im}(f)$. □

Recall that we have a bijection $\mathrm{Hom}_R(M \otimes_R N, L) \cong \mathrm{Bil}(M \times N, L)$ from the universal property of the tensor product. But

$$\mathrm{Bil}(M \times N, L) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, L))$$

and so we have an isomorphism

$$\mathrm{Hom}_R(M \otimes N, L) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, L))$$

sending φ to $n \mapsto (m \mapsto \varphi(m \otimes n))$

Lecture 7

Proposition 2.4.5. Let M be an R-module. Then T_M is a right exact functor.

Proof. Given an exact sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

Fix an R-module P. We will apply the functors $\mathrm{Hom}_R(\cdot, P)$, then the functor $\mathrm{Hom}_R(M, \cdot)$, to get the sequence

$$0 \longrightarrow \mathrm{Hom}_R(M, \mathrm{Hom}_R(C,P)) \longrightarrow \mathrm{Hom}_R(M, \mathrm{Hom}_R(B,P)) \longrightarrow \mathrm{Hom}_R(M, \mathrm{Hom}_R(A,P))$$

which is exact as the Hom functors are left exact. Using the isomorphism above, and noting that the square

$$\begin{array}{ccc} \mathrm{Hom}_R(M, \mathrm{Hom}_R(C, P)) & \longrightarrow & \mathrm{Hom}_R(M, \mathrm{Hom}_R(B, P)) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_R(M \otimes C, P) & \longrightarrow & \mathrm{Hom}_R(M \otimes B, P) \end{array}$$

commutes, we have an exact sequence

$$\mathrm{Hom}_R(M \otimes C, P) \longrightarrow \mathrm{Hom}(M \otimes B, P) \longrightarrow \mathrm{Hom}(M \otimes A, P) \longrightarrow 0$$

Since P is arbitrary, using lemma 2.4.4, we see that

$$\mathrm{T}_M(A) \longrightarrow \mathrm{T}_M(B) \longrightarrow \mathrm{T}_M(C) \longrightarrow 0$$

is exact, as required. □

Remark 2.4.6. Note on the other hand that

$$A \longrightarrow B \longrightarrow C$$

being exact does not imply that

$$\mathrm{T}_M(A) \longrightarrow \mathrm{T}_M(B) \longrightarrow \mathrm{T}_M(C)$$

is exact.

For example, consider the exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}$$

This is exact, but

$$0 \longrightarrow \mathbb{Z} \otimes \mathbb{Z}/2 \xrightarrow{\cdot 2} \mathbb{Z} \otimes \mathbb{Z}/2$$

is not.

2.5 Flat modules - a first encounter

Definition 2.5.1 (flat module)

An R-module M is *flat* if for any injective R-module homomorphism $N \to N'$, the map $\mathrm{T}_M(f) : \mathrm{T}_M(N) \to \mathrm{T}_M(N')$ is injective.

Example 2.5.2

$\mathbb{Z}/2$ is not a flat $\mathbb{Z}$-module, as seen in the remark above.

Example 2.5.3

Free modules are flat. To see this, suppose $f : N \to N'$ is an injective R-linear map. Then we have the commuting square

$$\begin{array}{ccc} R^{\oplus I} \otimes N & \xrightarrow{\mathrm{id} \otimes f} & R^{\oplus I} \otimes N' \\ \updownarrow & & \updownarrow \\ (R \otimes N)^{\oplus I} & & (R \otimes N')^{\oplus I} \\ \updownarrow & & \updownarrow \\ N^{\oplus I} & \xrightarrow{f^{\oplus I}} & (N')^{\oplus I} \end{array}$$

where the vertical maps are isomorphisms, and

$$f^{\oplus I}((n_i)_{i\in I}) = (f(n_i))_{i\in I}$$

It is clear that $f^{\oplus I}$ is injective.

Remark 2.5.4. With this, we see that the base ring matters. $\mathbb{Z}/2$ is not a flat $\mathbb{Z}$-module, but it is a flat $\mathbb{Z}/2$-module as it is free.

Definition 2.5.5 (torsion free)
An R-module is *torsion free* if for any $r \in R, m \in M$, $rm = 0$ implies that $m = 0$ or r is a zero divisor.

Proposition 2.5.6. Flat modules are torsion free.

Proof. Suppose M was not torsion free. Then there exists $r_0 \in R, m_0 \in M$ with r_0 not a zero divisor, $m_0 \neq 0$, such that $r_0 m_0 = 0$. We can define a map

$$f : R \to R$$
$$f(x) = r_0 x$$

f is injective as r_0 is not a zero divisor. Thus, we have the square

$$\begin{array}{ccc} M \otimes R & \xrightarrow{\mathrm{id}\otimes f} & M \otimes R \\ \updownarrow & & \updownarrow \\ M & \xrightarrow[r_0\cdot]{} & M \end{array}$$

But the bottom map is not injective, as it sends m_0 to zero. □

For a specical case of the above:

Proposition 2.5.7. Let R be an integral domain, I a non-zero, non-unit ideal. Then R/I is not flat.

Proof. Since $I \neq R$, R/I is non-zero. Choose $x \in I \setminus 0$, and consider the map

$$f : R \to R$$
$$f(r) = xr$$

This is an injective map. But the induced map on $R \otimes (R/I) \cong R/I$ is multiplication by x, which is the zero map. □

Proposition 2.5.8 (criterion for flatness)**.** Let M be an R-module. Then teh following are equivalent:

(i) T_M preserves exactness of all exact sequences,

(ii) T_M preserves exactness of short exact sequences,

(iii) T_M is flat,

(iv) if $f : N \to N'$ is R-linear and injective, N, N' are finitely generated R-modules, then $\mathrm{id}_M \otimes f$ is injective.

Proof. $(i) \implies (ii) \implies (iii) \implies (iv)$ is clear.

For $(ii) \implies (i)$, suppose

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact, then we have a short exact sequence

$$0 \longrightarrow \frac{A}{\ker(f)} \xrightarrow{\overline{f}} B \xrightarrow{g} \operatorname{im}(g) \longrightarrow 0$$

Thus, we have a short exact sequence

$$0 \longrightarrow M \otimes \frac{A}{\ker(f)} \longrightarrow M \otimes B \longrightarrow M \otimes \operatorname{im}(g) \longrightarrow 0$$

That is, $\ker(\mathrm{id}_M \otimes g) = \operatorname{im}(\mathrm{id}_M \otimes \overline{f}) = \operatorname{im}(\mathrm{id}_M \otimes f)$. Thus the sequence

$$M \otimes A \longrightarrow M \otimes B \longrightarrow M \otimes C$$

is exact.

We will omit the proof of $(iv) \implies (iii)$, it can be found in the lecturer's notes.

For $(iii) \implies (ii)$, we note that this follows from T_M being right exact. □

Proposition 2.5.9. Let $f : R \to S$ be a ring homomorphism, M is a flat R-module. Then $S \otimes_R M$ is a flat S-module.

Proof. Let $g : N \to N'$ be an injective S-linear map. Then the square

$$\begin{array}{ccc} (S \otimes_R M) \otimes_S N & \longrightarrow & (S \otimes_R M) \otimes_S N' \\ \updownarrow & & \updownarrow \\ M \otimes_R N & \longrightarrow & M \otimes_R N' \end{array}$$

commutes. But the bottom map is injective as M is flat. □

Lecture 8

2.6 Further examples of tensor products

Example 2.6.1

First consider $x \otimes y \in \mathbb{Q} \otimes_Z (\mathbb{Z}/n)$. We can write

$$x \otimes y = n\frac{x}{n} \otimes y = \frac{x}{n} \otimes ny = 0$$

and so, $\mathbb{Q} \otimes_{\mathbb{Z}} (\mathbb{Z}/n) = 0$. We used the fact that $\mathbb{Q}$ is a *divisible group*, that is, for all $x \in \mathbb{Q}, n \in \mathbb{N}$, there exists $y \in \mathbb{Q}$ such that $ny = x$. Moreover, we also used the fact that $\mathbb{Z}/n$ is torsion.

More generally,

$$\text{divisible} \otimes \text{torsion} = 0$$

and so

$$(\mathbb{Q}/\mathbb{Z}) \otimes_Z (\mathbb{Q}/\mathbb{Z})$$

But for an R-module M which is non-zero, if M is finitely generated, then $M \otimes_R M \neq 0$.

Example 2.6.2

Let V be a $\mathbb{Q}$ vector space, then

$$\mathbb{Q} \otimes_{\mathbb{Q}} V = V$$

But in this case, we also have that

$$\mathbb{Q} \otimes_{\mathbb{Z}} V = V$$

with $x \otimes v \mapsto xv$.

Proof. Every tensor in $\mathbb{Q} \otimes_{\mathbb{Z}} V$ is pure, since we can write

$$\sum \frac{a_i}{b_i} \otimes v_i = \sum \frac{1}{b_i} \otimes (a_i v_i) = \sum \frac{1}{b_i} \otimes \frac{a_i}{b_i} v_i = \sum 1 \otimes \frac{a_i}{b_i} v_i = 1 \otimes \sum \frac{a_i}{b_i}$$

Clearly this map is surjective, and it is easy to see that if $xv = 0$ then either $x = 0$ or $v = 0$. □

Example 2.6.3

Recall that

$$M \otimes_R \left(\bigoplus_{i \in I} N_i \right) \cong \bigoplus_{i \in I} (M \otimes N_i)$$

On the other hand, if we consider the direct product, we have a map

$$\begin{aligned} M \otimes \prod_i N_i &\to \prod_i (M \otimes N_i) \\ m \otimes (n_i) &\mapsto (m \otimes n_i) \end{aligned}$$

which is in general, not an isomorphism. For example, consider

$$\mathbb{Q} \otimes_{\mathbb{Z}} \prod_{n \geq 1} \frac{\mathbb{Z}}{2^n} \to \prod_{n \geq 1} \mathbb{Q} \otimes \frac{\mathbb{Z}}{2^n}$$

But from above, $\mathbb{Q} \otimes (\mathbb{Z}/2^n) = 0$, and so the right hand side is zero. For the left hand side, take

$$g = (1, 1, \dots) \in \prod_{n \geq 1} \frac{\mathbb{Z}}{2^n}$$

Note that g has infinite order, and so it generates a subgroup isomorphic to Z. But recall that

$$\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z} = \mathbb{Q}$$

With this, we have an injective map

$$\mathbb{Q} \otimes \langle g \rangle \hookrightarrow \mathbb{Q} \otimes \prod_{n \geq 1} \frac{\mathbb{Z}}{2^n}$$

We will see later that $\mathbb{Q}$ is a flat $\mathbb{Z}$-module.

Example 2.6.4

Consider $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ as an $\mathbb{C}$-algebra, where we first restrict scalars on the right copy of $\mathbb{C}$, and extend scalars using the left copy.

Recall that as a $\mathbb{C}$-vector space,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^2 \cong \mathbb{C}^2$$

and we have a basis $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$, which is $1 \otimes 1, 1 \otimes i$ as a $\mathbb{C}$-vector space.

To consider this as a $\mathbb{C}$-algebra, then

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \otimes_{\mathbb{R}} \frac{\mathbb{R}[t]}{\langle t^2 + 1 \rangle} \cong \frac{\mathbb{C}[t]}{\langle t^2 + 1 \rangle} = \frac{\mathbb{C}[t]}{\langle t - i \rangle \langle t + i \rangle} \cong \frac{\mathbb{C}[t]}{\langle t - i \rangle} \times \frac{\mathbb{C}[t]}{\langle t + i \rangle} \cong \mathbb{C} \times \mathbb{C}$$

where we used the Chinese remainder theorem. On a pure tensor, we have

$$(a+bi)\otimes(c+di)\mapsto(a+bi)\otimes\underbrace{[c+dt]}_{\text{coset of } c+dt}\mapsto(a+bi)[c+dt]$$

We can compute this, to get

$$P=(ac+bdit)+(ibc+tad)$$

and we then have

$$P\mapsto(ac-bd+i(bc+ad),ac+bd+i(bc-ad))$$

If we set $x=a+bi, y=c+di$, then the result is just $(xy, x\overline{y})$.

3 Localisation

Definition 3.0.1 (multiplicative subset)

A *multiplicative(ly closed) subset* $S\subseteq\mathbb{R}$ such that

1. $1\in S$,
2. if $a,b\in S$, then $ab\in S$.

If $U\subseteq R$ is any set, then the *multiplicative closure* S of U is the set of

$$\prod_{i=1}^{n}u_i$$

where $u_i\in U$, $n\geq 0$.

Example 3.0.2

If R is an integral domain, then $S=R\setminus\{0\}$ is multiplicative. More generally, if $\mathfrak{p}\trianglelefteq R$ is a prime ideal (of any ring R), then $S=R\setminus\mathfrak{p}$ is multiplicative.

Example 3.0.3

If $x\in R$, then $S=\{1,x,x^2,\dots\}$ is multiplicative.

Example 3.0.4

$\mathbb{Q}$ is obtained from $\mathbb{Z}$ by adding inverses for the elements of the multiplicative subset $\mathbb{Z}\setminus\{0\}$, and we have a ring homomorphism $\mathbb{Z}\hookrightarrow\mathbb{Q}$.

We will generalise this example to general rings R, and with arbitrary multiplicative subsets $S\subseteq R$. But in general, we will lose injectivity.

Lecture 9

3.1 Construction

Definition 3.1.1 (localisation)

Let $S\subseteq R$ be a multiplicative set, M is an R-module. Consider the set $M\times S$, with the relation $(m_1,s_1)\sim(m_2,s_2)$ if there exists $u\in S$, such that

$$u(s_2m_1-s_1m_2)=0$$

This is an equivalence relation, and we $S^{-1}M$ for the set of equivalence classes. We write

$$\frac{m}{s} = [(m, s)]$$

for the equivalence class. Finally, we write

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} = \frac{m_1 s_2 + m_2 s_1}{s_1 s_2}$$

and

$$r \cdot \frac{m}{s} = \frac{rm}{s}$$

The above makes $S^{-1}M$ into an R-module. We call $S^{-1}M$ the *localisation of M at S*.

If $M = R$, we can make $S^{-1}R$ into a ring by

$$\frac{r_1}{s_1} \cdot \frac{r_2}{s_2}$$

Next, we note that we have an $S^{-1}R$-module structure on $S^{-1}M$, via

$$\frac{r}{s} \cdot \frac{m}{t} = \frac{rm}{st}$$

We have localisation maps:

$$R \to S^{-1}R$$
$$r \mapsto \frac{r}{1}$$

which is a ring homomorphism, and

$$M \to S[-1]M$$
$$m \mapsto \frac{m}{1}$$

which is an R-linear map.

We check that $\sim$ above defines an equivalence relation: Reflexivity and symmetry are clear. Say $(m_1, s_1) \sim (m_2, s_2)$ and $(m_2, s_2) \sim (m_3, s_3)$. That is, there exists $u, v \in S$ such that

$$u(s_2 m_1 - s_1 m_2) = v(s_3 m_2 - s_2 m_3) = 0$$

Multiplying the first term by vs_3 and the second by us_1, we get

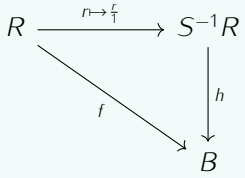
$$uvs_2 s_3 m_1 = uvs_3 s_1 m_2$$
$$uvs_1 s_3 m_2 = uvs_1 s_2 m_3$$

and so, we have that

$$uvs_2(s_3 m_1 - s_1 m_3) = 0$$

Since S is multiplicatively closed, we are done.

Proposition 3.1.2 (universal property of $S^{-1}R$)**.** Let $U \subseteq R$ be any subset, and let $S \subseteq R$ be the multiplicative closure of U. Let $f : R \to B$ be a ring homomorphism, such that $f(u)$ is a unit for all $u \in U$. Then there exists a unique ring homomorphism $h : S^{-1}R \to B$, such that the diagram



commutes. That is,

$$f(r) = h\left(\frac{r}{1}\right)$$

Another way of thinking about this is that we have a bijection

$$\mathrm{Hom}_{\mathrm{Ring}}(S^{-1}R, B) \leftrightarrow \{\varphi : R \to B \text{ ring hom., with } \varphi(U) \subseteq B^\times\}$$

given by sending f to $r \mapsto f\left(\frac{r}{1}\right)$.

Proof. Let $f : R \to B$ be a ring homomorphism, with $f(U) \subseteq B^\times$. In this case, $f(S) \subseteq B^\times$ as well. We want $h : S^{-1}R \to B$, with

$$f(r) = h\left(\frac{r}{1}\right)$$

First, such h must satisfy:

$$1 = h(1) = h\left(\frac{1}{s} \cdot \frac{s}{1}\right) = h\left(\frac{1}{s}\right) f(s)$$

Thus, we must have that $h(1/s) = f(s)^{-1}$. With this, we have

$$h\left(\frac{r}{s}\right) = h\left(\frac{r}{1}\right) h\left(\frac{1}{s}\right) = f(r)f(s)^{-1}$$

But we need to check if h is well defined. That is, if $r_1/s_1 = r_2/s_2$, then there exists $t \in S$ such that $t(s_2r_1 - s_1r_2) = 0$, or equivalently,

$$ts_2r_1 = ts_1r_2$$

Applying f, we get

$$f(t)f(s_2)f(r_1) = f(t)f(s_1)f(r_2)$$

But every element in the above equality are in $B^\times$, and so we are done. It is easy to check that h is a ring homomorphism. □

Proposition 3.1.3. If (A, j) satisfies the same universal property of $(S^{-1}R, \iota)$, where $\iota(r) = r/1$, then there exists an isomorphism $S^{-1}R \to A$, sending

$$\frac{r}{s} \mapsto j(r)j(s)^{-1}$$

Facts

1. Take $r/s \in S^{-1}R$, then
$$\frac{r}{s} = \frac{0}{1} \iff \text{there exists } u \in S \text{ with } ur = 0$$
2. $S^{-1}R = 0$ if and only if $0 \in S$.
3. $$\ker(\iota : R \to S^{-1}R) = \{r \in R \mid \text{ there exists } u \in S \text{ with } ur = 0\}$$
4. In particular, ι is injective if and only if S does not contain any zero divisors.
5. ι is always an epimorphism[1], but usually not surjective. For example, $\iota : \mathbb{Z} \to \mathbb{Q}$ is an epimorphism. If we have $f, g : \mathbb{Q} \to A$ ring homomorphisms, with $f \circ \iota = g \circ \iota$, then $f = g$.

Example 3.1.4

For $f \in R$, let $S = \{f^n \mid n \geq 0\}$. Then we define $R_f = S^{-1}R$.

[1] A morphism $f : X \to Y$ (in some category) is called an *epimorphism* if for all $g_1, g_2 : Y \to Z$, with $g_1 \circ f = g_2 \circ f$, we have $g_1 = g_2$.

If $R = \mathbb{Z}, f = 2$, then

$$R_f = \left\{ \frac{a}{2^n} \mid a \in \mathbb{Z}, n \geq 0 \right\} = \mathbb{Z}\left[\frac{1}{2}\right]$$

Notation 3.1.5. In this course, we will write:

- $\mathbb{Z}/n$ for the finite ring,
- $\mathbb{Z}_2$ for the 2-adic integers,
- $\mathbb{Z}[1/2]$ for the above ring.

Example 3.1.6

For a ring R, let $\operatorname{Spec}(R)$ denote its prime spectrum. For $\mathfrak{p} \in \operatorname{Spec}(R)$, we can let $S = R \setminus \mathfrak{p}$, and we write $R_{\mathfrak{p}} = (R \setminus \mathfrak{p})^{-1}R$.

If $R = \mathbb{Z}$, $p = \langle 3 \rangle$, then

$$\mathbb{Z}_{\langle 3 \rangle} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, 3 \nmid b \right\}$$

Proposition 3.1.7. If M is an R-module, $S \subseteq R$ a multiplicative subset, then we have an isomorphism:

$$\begin{aligned} \underset{R}{\otimes} M &\to S^{-1}M \\ \frac{r}{s} \otimes m &\mapsto \frac{rm}{s} \end{aligned}$$

Proof. We can define a bilinear map

$$\begin{aligned} S^{-1}R \times M &\to S^{-1}M \\ \left(\frac{r}{s}, m\right) &\mapsto \frac{rm}{s} \end{aligned}$$

and thus, by the universal property we haev $\varphi : S^{-1}R \otimes_R M \to S^{-1}M$. This is R-linear, and it is easy to see that φ is also $S^{-1}R$-linear. It is clear that φ is surjective, since

$$\varphi\left(\frac{1}{s} \otimes m\right) = \frac{m}{s}$$

We want to show that every tensor

$$t = \sum_i \frac{r_i}{s_i} \otimes m_i \in S^{-1}R \otimes_R M$$

is prime. Define $s = \prod_i s_i$, and $t_j = \prod_{i \neq j} s_i$. In this case,

$$\begin{aligned} \sum \frac{r_i}{s_i} \otimes m_i &= \sum \frac{1}{s_i} \otimes (r_i m_i) \\ &= \sum \frac{t_i}{s} \otimes (r_i m_i) \\ &= \frac{1}{s} \otimes \left(\sum_i r_i t_i m_i \right) \end{aligned}$$

Using this, if

$$\varphi\left(\frac{1}{s} \otimes m\right) = \frac{m}{s} = 0 = \frac{0}{1}$$

That is, there exists $u \in S$, such that $um = 0$. In this case,

$$\frac{1}{s} \otimes m = \frac{u}{us} \otimes m = \frac{1}{us} \otimes (um) = 0$$

□

With this, $S^{-1}R \otimes (\cdots)$ acts on R-modules. But in fact, it also acts on R-linear maps.

Proposition 3.1.8 (localisation is a functor)**.** Let M be an R-module, $S \subseteq R$ a multiplicative subset. Let $f : N \to N'$ be an R-linear map. Then the following square commutes:

$$\begin{array}{ccc} S^{-1}R \otimes N & \xrightarrow{\mathrm{id}_{S^{-1}R} \otimes f} & S^{-1}R \otimes N' \\ \downarrow \sim & & \downarrow \sim \\ S^{-1}N & \xrightarrow[S^{-1}(f)]{} & S^{-1}N' \end{array}$$

In particular,

$$(S^{-1}f)\left(\frac{n}{s}\right) = \frac{f(n)}{s}$$

With this, the functors $S^{-1}R \otimes (\cdot)$ and $S^{-1}(\cdot)$ are naturally isomorphic.

Remark 3.1.9. Let A be an R-algebra, $S^{-1}R \otimes A \to S^{-1}A$ is $S^{-1}R$-linear, and also an isomorphism of $S^{-1}R$-algebras.

Lemma 3.1.10. If M is an $S^{-1}R$-module, then, we can restrict scalars on M from $S^{-1}R$ to R, then apply $S^{-1}(\cdot)$. Then

$$S^{-1}M \cong M$$

as $S^{-1}R$-modules. Equivalently,

$$M \cong S^{-1}R \otimes M$$

as $S^{-1}R$-modules.

Proof. We can see that the map

$$M \to S^{-1}M \tag{1}$$

$$m \mapsto \frac{m}{1} \tag{2}$$

is $S^{-1}R$-linear. Surjectivity and injectivity are clear. □

Proposition 3.1.11. Let M be an R-module, L an $S^{-1}R$-module, $f : M \to L$ is R-linear. Then there exists a unique $h : S^{-1}M \to L$ which is $S^{-1}R$-linear, such that

$$f(m) = h\left(\frac{m}{1}\right)$$

Proof. We know that $S^{-1}(\cdot) \otimes S^{-1}R \otimes (\cdot)$, and so it suffices to prove the result for the tensor product. With this, the localisation map is

$$\iota : M \to S^{-1}R \otimes M$$

$$m \mapsto \frac{1}{1} \otimes m$$

Let $f : M \to L$ be R-linear. We then have that

$$h : \mathrm{id}_{S^{-1}R} \otimes f : S^{-1}R \otimes_R M \to S^{-1}R \otimes_R L$$

But the previous lemma shows that $S^{-1}R \otimes_R L \cong L$ as $S^{-1}R$-modules. In particular,

$$h\left(\frac{r}{s} \otimes m\right) = \frac{r}{s} f(m)$$

For the uniqueness of h, it follows from the fact that elements of the form $\frac{1}{1} \otimes m$ generate $S^{-1}R \otimes_R M$ as an $S^{-1}R$-module. □

Proposition 3.1.12 (the functor $S^{-1}R$ is exact)**.** If

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is an exact sequence of R-modules, then

$$S^{-1}A \xrightarrow{S^{-1}f} S^{-1}B \xrightarrow{S^{-1}g} S^{-1}C$$

is an exact sequence of $S^{-1}R$-modules.

Proof.

$$(S^{-1}g) \circ (S^{-1}f) = S^{-1}(g \circ f) = S^{-1}(0) = 0$$

and so $\operatorname{im}(S^{-1}f) \subseteq \ker(S^{-1}g)$. Let

$$\frac{b}{s} \in \ker(S^{-1}g)$$

Then

$$\frac{g(b)}{s} = \frac{0}{1}$$

That is, there exists $u \in S$, such that $u \cdot g(b) = 0$. But g is R-linear, $u \in R$, and so $g(ub) = 0$, which means that $ub \in \ker(g) = \operatorname{im}(f)$. Thus, there exists $a \in A$ such that $f(a) = ub$. Now

$$\frac{b}{s} = \frac{ub}{us} = \frac{f(a)}{us} = S^{-1}f\left(\frac{a}{us}\right) \in \operatorname{im}(S^{-1}f)$$

□

Equivalently, $S^{-1}R$ is a flat R-module. Suppose $\iota : N \to M$ is the inclusion map, then

$$S^{-1}\iota : S^{-1}N \to S^{-1}M$$

is injective, and so the expression

$$\frac{n}{s}$$

makes sense in $S^{-1}N$ *and* $S^{-1}(M)$.

Proposition 3.1.13. Let M be an R-module, N, P submodules of M. Then

(i) $S^{-1}(N + P) = S^{-1}N + S^{-1}P$.

(ii) $S^{-1}(N \cap P) = S^{-1}N \cap S^{-1}P$,

(iii) $(S^{-1}M)/(S^{-1}N) \cong S^{-1}(M/N)$ as $S^{-1}R$ modules via

$$\frac{m}{s} + S^{-1}N \leftrightarrow \frac{m + N}{s}$$

Proof. For (i), the left hand side consists of elements of the form $\frac{n+p}{s}$, and the right hand side consists of elements of the form $\frac{n}{s_1} + \frac{p}{s_2}$. The result is then clear.

For (ii), $\subseteq$ is clear. Given $x \in S^{-1}N \cap S^{-1}P$, that is,

$$x = \frac{n}{s_1} = \frac{p}{s_2}$$

for $n \in N, p \in P, s_1, s_2 \in S$. But then there exists $u \in S$, such that $us_2n = us_1p =: w \in N \cap P$. With this,

$$x - \frac{n}{s_1} = \frac{us_2n}{us_1s_2} = \frac{w}{us_1s_2} \in S^{-1}(N \cap P)$$

For (iii), consider the exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0$$

Applying the exact functor S^{-1},

$$0 \longrightarrow S^{-1}N \longrightarrow S^{-1}M \longrightarrow S^{-1}(M/N) \longrightarrow 0$$

But this immediately gives that

$$S^{-1}(M/N) \cong \frac{S^{-1}M}{S^{-1}N}$$

as $S^{-1}R$-modules. Computing the respective maps gives the result. □

Proposition 3.1.14. If M, N are R-modules, then

$$S^{-1}M \otimes_{S^{-1}R} S^{-1}N \cong S^{-1}(M \otimes_R N)$$

Proof. We have the isomorphism from extension of scalars:

$$(S^{-1}R \otimes_R M) \otimes_{S^{-1}R} (S^{-1}R \otimes_R N) \cong S^{-1}R \otimes_R (M \otimes_R N)$$

□

A special case of this is that if $\mathfrak{p}$ is a prime ideal of R, then

$$M_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} N_{\mathfrak{p}} = (M \otimes_R N)_{\mathfrak{p}}$$

3.2 Extension and contraction of ideals

Recall if $f : A \to B$ is a ring homomorphism, we define the *contraction of* $\mathfrak{b} \trianglelefteq B$ as

$$\mathfrak{b}^c = f^{-1}(\mathfrak{b}) \trianglelefteq A$$

and the *extension of* $\mathfrak{a} \trianglelefteq A$ as

$$\mathfrak{a}^e = \langle f(\mathfrak{a}) \rangle \trianglelefteq B$$

In examples sheet 1, we have a bijection

$$\{\text{contracted ideals of } A\} \leftrightarrow \{\text{extended ideals of } B\}$$

To see this, we have that an ideal $\mathfrak{a}$ is contracted if and only if $\mathfrak{a} = \mathfrak{a}^{ec}$, and an ideal $\mathfrak{b}$ is extended if and only if $\mathfrak{b} = \mathfrak{b}^{ce}$, and so the bijection is given by extension/contraction.

Let S be a multiplicative subset of R, and we will consider the ring homomorphism $R \to S^{-1}R$, given by $r \mapsto r/1$. For an ideal $\mathfrak{a}$ of R, we have the *extension*

$$\mathfrak{a}^e = S^{-1}\mathfrak{a} \trianglelefteq S^{-1}R$$

and for an ideal $\mathfrak{b}$ of $S^{-1}R$, we have the contraction $\mathfrak{b}^c \trianglelefteq R$.

Proposition 3.2.1.

$$\mathfrak{a}^e = S^{-1}\mathfrak{a} = \left\{ \frac{a}{s} \;\middle|\; a \in \mathfrak{a}, s \in S \right\}$$

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